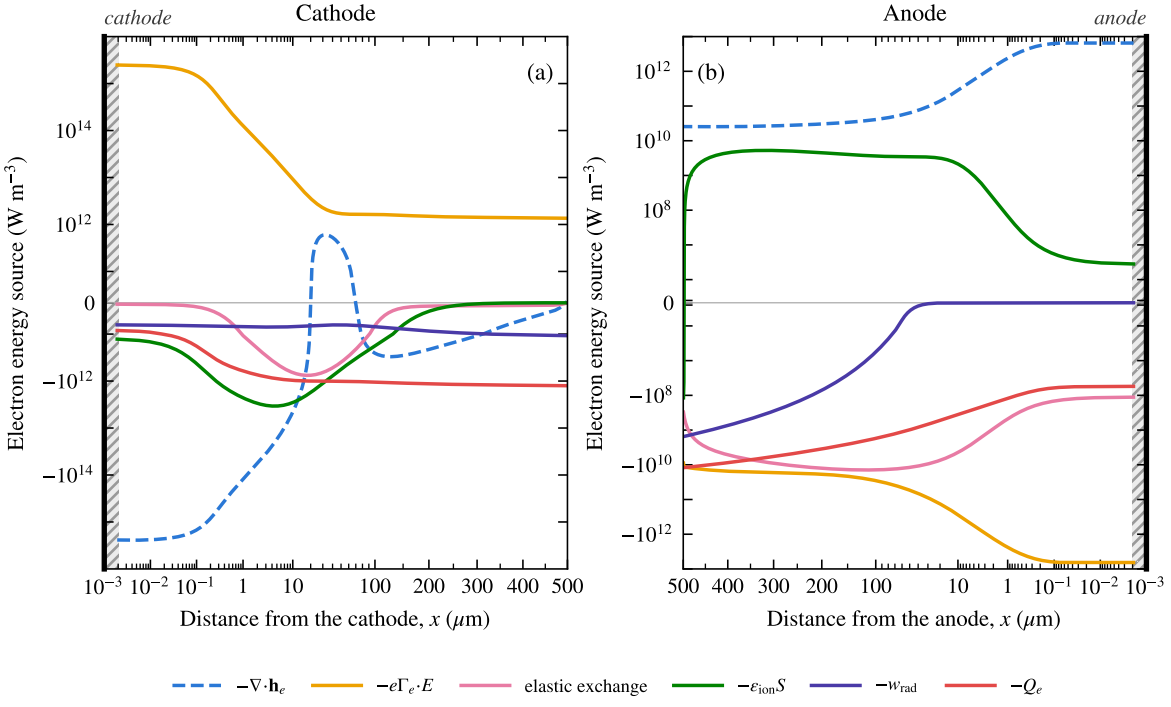
cathode
Cathode
(a)
Electron energy source (W m$^{-3}$)
$10^{14}$
$10^{12}$
0
$-10^{12}$
$-10^{14}$
$10^{-3}$ $10^{-2}$ $10^{-1}$ 1 10 100 200 300 400 500
Distance from the cathode, $x$ ($\mu$m)
Anode
anode
(b)
Electron energy source (W m$^{-3}$)
$10^{12}$
$10^{10}$
$10^{8}$
0
$-10^{8}$
$-10^{10}$
$-10^{12}$
500 400 300 200 100 10 1 $10^{-1}$ $10^{-2}$ $10^{-3}$
Distance from the anode, $x$ ($\mu$m)
$-\nabla\cdot\mathbf{h}_e$
$-e\Gamma_e\cdot E$
elastic exchange
$-\varepsilon_{\text{ion}}S$
$-w_{\text{rad}}$
$-Q_e$